FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

ASYNCHRONOUS ACTIVITY

1

Uploading a KNIME Project to GitHub

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STUDENT GUIDE

Uploading a KNIME Project to GitHub

Complete step by step instructions using Git Bash

Prepared for students submitting exported KNIME workflows

Final repository structure	Required contents
Project folder	Exported KNIME workflow file
README.md	Project description and instructions
Dataset	CSV file required by the workflow

CS0065 | Machine Learning Workflow Submission

Purpose and Expected Output

This guide explains how to prepare an exported KNIME workflow, place its supporting files in one project folder, document the project with a README file, and upload the complete project to a GitHub repository. After completing the procedure, students must verify that the workflow, dataset, and README are visible online and submit the repository link.

Before You Begin

Requirement	What students need
GitHub account	An account using the required institutional or approved email address
Git for Windows	Git Bash installed and available on the computer
KNIME workflow	An exported .knwf workflow file
Dataset	The CSV or other source file used by the workflow
Text editor	Notepad, Notepad++, or Visual Studio Code for README.md
Internet access	Required to authenticate and push files to GitHub

Important Terms

Term	Meaning
Repository	The project storage location on GitHub
Commit	A recorded version of the files with a short description
Push	Sending committed files from the computer to GitHub
Pull	Retrieving and combining changes already stored on GitHub
README.md	A Markdown text file that describes the project

Important: GitHub does not track empty folders. Every submitted project folder must contain its actual files.

Part 1 Prepare the Project Folder

1. Create a folder for the project. Use a clear name without unnecessary special characters. Example: DemoEarlyWarningTool.
2. Place the exported KNIME workflow in the folder. The filename should end in .knwf.
3. Place the dataset used by the workflow in the same folder. Example: student_performance_knime.csv.
4. Create README.md and place it in the same folder.

Recommended local structure:

```
DemoEarlyWarningTool/
  DemoEarlyWarningTool.knwf
```

```
README.md
student_performance_knime.csv
```

Part 2 Create README.md Using Notepad

5. Open Notepad.
6. Type or paste the project documentation.
7. Select File, then Save As.
8. Choose the project folder as the save location.
9. Enter README.md in the File name box.
10. Set Save as type to All files (*.*).
11. Set Encoding to UTF-8, then select Save.

Check: The filename must be README.md, not README.md.txt. In File Explorer, enable View > Show > File name extensions if necessary.

Suggested README Content

```
# Asynchronous Activity 1: Student Early Warning Tool Using KNIME

This repository contains the required output for Asynchronous Activity 1. The project uses
KNIME to predict student risk status from academic performance data.

## Files
- `DemoEarlyWarningTool.knwf` - exported KNIME workflow
- `student_performance_knime.csv` - dataset used by the workflow

## Algorithms
- Decision Tree
- Logistic Regression
- Random Forest

## How to Run
1. Import the `.knwf` file into KNIME Analytics Platform.
2. Configure the CSV Reader and select the included dataset.
3. Execute all nodes.
4. Open the Scorer nodes to review the evaluation results.

## Author
[Student name]

## Course and Section
[Course and section]
```

Part 3 Create the GitHub Repository

12. Sign in to GitHub.
13. Select the plus (+) icon in the upper-right corner, then select New repository.
14. Enter the repository name required by your instructor. Follow the prescribed naming convention.
15. Add a short description, if required.
16. Choose the required visibility: Public or Private.
17. Select Create repository.

Record: Copy the repository URL. It normally follows this pattern:
https://github.com/USERNAME/REPOSITORY-NAME.git

Part 4 Open the Correct Folder in Git Bash

Open the project folder in File Explorer. Right-click an empty area and select Open Git Bash here. If that option is unavailable, open Git Bash and navigate using cd.

Paths With Spaces

Git Bash uses forward slashes. Place a path containing spaces inside quotation marks.

```
cd "/d/FEU ALABANG/COURSES/TERM 1 SY2026-2027/CS0065/MACHINE LEARNING
ALGO/DemoEarlyWarningTool"
```

Confirm the current location and display the files:

```
pwd
ls
```

Expected result: The ls command should display the .knwf file, README.md, and dataset.

Common Navigation Errors

Message	Cause	Correction
cd: too many arguments	The path contains spaces and was not quoted	Put the complete path inside double quotation marks
cd: No such file or directory	The folder name or path is incorrect	Run ls and use the exact folder name
command not found	A command was misspelled, such as get or ~git	Type git exactly, without a tilde

Part 5 Configure Git Identity

Perform this configuration once on the computer. Use the name and email associated with the student's GitHub account.

```
git config --global user.name "Student Name"
git config --global user.email "student@example.com"
```

Verify the saved values:

```
git config --global user.name
git config --global user.email
```

Important: The Git identity identifies the author of a commit. It does not grant permission to another person's repository.

Part 6 Initialize and Commit the Project

18. Initialize the current project folder as a local Git repository.

```
git init
```

19. Rename the current branch to main.

```
git branch -M main
```

20. Check the repository status.

```
git status
```

21. Stage all files in the current project folder. The period means all current files and folders.

```
git add .
```

22. Check that the files appear under Changes to be committed.

```
git status
```

23. Create the first commit. Keep the complete message inside quotation marks.

```
git commit -m "Add KNIME workflow and project files"
```

Part 7 Connect the GitHub Repository

Replace USERNAME and REPOSITORY-NAME with the actual GitHub values.

```
git remote add origin https://github.com/USERNAME/REPOSITORY-NAME.git
```

Verify the remote repository:

```
git remote -v
```

If origin already exists: Do not add it again. Verify it with git remote -v. If it is incorrect, ask the instructor before changing it.

Part 8 Push the Project to GitHub

```
git push -u origin main
```

A browser may open for GitHub authentication. Sign in using the account that owns the repository or an account that has collaborator access. After a successful push, refresh the repository page on GitHub.

Part 9 If the GitHub Repository Already Contains Files

A push may be rejected as non-fast-forward when the online repository already contains a README, notebook, or another commit. Retrieve and merge the online history first:

```
git pull origin main --allow-unrelated-histories --no-rebase --no-edit
```

If the pull completes without a conflict, push again:

```
git push -u origin main
```

If README.md Has a Merge Conflict

If the local README is the version that should be submitted, keep it and complete the merge:

```
git checkout --ours README.md
git add README.md
git commit -m "Merge remote repository with KNIME project"
git push -u origin main
```

Safety: Do not use git push --force for this activity. A normal merge preserves existing repository work.

Part 10 Organize the Files in a Project Folder on GitHub

When Git is initialized inside DemoEarlyWarningTool, GitHub displays its files at the repository root. To display an actual DemoEarlyWarningTool folder inside a repository that also contains other projects, create the folder and move the tracked files into it:

```
mkdir -p DemoEarlyWarningTool
git mv DemoEarlyWarningTool.knwf DemoEarlyWarningTool/
git mv student_performance_knime.csv DemoEarlyWarningTool/
git mv README.md DemoEarlyWarningTool/
git add .
git commit -m "Organize KNIME files into project folder"
git push
```

Expected GitHub structure:

```
REPOSITORY-NAME/
  DemoEarlyWarningTool/
    DemoEarlyWarningTool.knwf
    README.md
    student_performance_knime.csv
```

README behavior: A README.md inside the project folder is displayed when that folder is opened on GitHub.

Part 11 Update the Project Later

After editing the KNIME workflow, README, or dataset, open Git Bash in the local repository folder and run:

```
git status
git add .
git commit -m "Update KNIME workflow"
git push
```

Use a clear commit message that describes the change. Examples include Update dataset, Correct README instructions, and Add Random Forest results.

Troubleshooting Guide

Error or symptom	Meaning	Required action
fatal: not a git repository	The current folder has not been initialized, or Git Bash is in the wrong folder	Navigate to the project folder and run git init
pathspec did not match any files	The named file or folder is absent from the current location	Run ls. If already inside the project folder, use git add .
Author identity unknown	Git name and email are not configured	Set user.name and user.email, then repeat the commit
switch m requires a value	The commit message after -m is missing or malformed	Use git commit -m "Complete message"
403 Permission denied	The authenticated account does not own or cannot write to the repository	Sign in as the owner or accept a collaborator invitation
non-fast-forward rejection	GitHub contains commits missing from the local branch	Pull and merge the remote history, then push again
README.md.txt appears	Notepad added .txt to the filename	Save as All files and use the exact name README.md

GitHub Authentication and Permission

If Git reports that access is denied to a different username, Windows may have saved another GitHub account. Remove the github.com entry from Windows Credential Manager, repeat the push, and authenticate with the correct account. Alternatively, the repository owner may add the student as a collaborator under Repository Settings and Collaborators.

Part 12 Prepare the Evidence PDF and Submit in MS Teams

After completing all GitHub instructions, students must prepare one PDF file containing screenshots that show the steps they completed. This PDF serves as proof that the project folder was prepared, committed, pushed, and verified successfully.

24. Take clear screenshots while completing the activity. Include the project folder and required files, the important Git Bash commands and successful results, and the final GitHub repository showing the project folder.
25. Arrange the screenshots in the same order as the steps in this guide. Add a short caption below each screenshot explaining what was completed.
26. Place the screenshots in Microsoft Word, Google Docs, or another document editor, then export or download the document as a PDF.
27. Name the evidence file using this format: Surname_FirstName_KNIME_GitHub_Evidence.pdf.
28. Copy the completed evidence PDF into the DemoEarlyWarningTool project folder together with the KNIME workflow, README.md, and dataset.
29. Push the evidence PDF and any final project changes to GitHub using the commands shown below.
30. Open the repository in a browser and verify that the DemoEarlyWarningTool folder contains all required files and that the evidence PDF opens correctly.
31. Copy the GitHub repository link. The submitted link must open the repository that contains the DemoEarlyWarningTool folder.
32. Open the assigned activity in Microsoft Teams, paste or attach the GitHub repository link in the submission area, and select Turn in.

Commands for the Final Push

```
git status
git add .
git commit -m "Add step-by-step evidence PDF"
git push
```

Required Final GitHub Structure

Location	Required file or folder
Repository root	DemoEarlyWarningTool folder
DemoEarlyWarningTool	DemoEarlyWarningTool.knwf
DemoEarlyWarningTool	README.md
DemoEarlyWarningTool	student_performance_knime.csv
DemoEarlyWarningTool	Surname_FirstName_KNIME_GitHub_Evidence.pdf

MS Teams Submission Requirement

Submit only after verification: The MS Teams assignment submission must contain the GitHub repository link that leads to the repository containing the DemoEarlyWarningTool folder. A local computer path, screenshot alone, or link to an individual file is not an acceptable substitute for the repository link.

Final Verification and Submission Checklist

Students must verify the online repository before submitting the activity. Do not submit only a screenshot of Git Bash.

☐ The GitHub repository opens using the submitted link.

☐ The correct project folder is visible.

☐ The exported KNIME workflow file ends in . knwf.

☐ README.md is present and is not saved as README.md.txt.

☐ The required dataset is present.

☐ The README identifies the project, workflow files, algorithms, and instructions.

☐ The most recent commit is visible on GitHub.

☐ The workflow can be downloaded and imported into KNIME.

☐ The CSV Reader can be redirected to the included dataset.

☐ The repository link follows the instructor's submission requirements.

☐ A PDF containing screenshots of the completed steps is included in the project folder.

☐ The evidence PDF opens correctly from the GitHub repository.

☐ The repository link was submitted through the assigned MS Teams activity.

Information to Submit

Required field	Student entry
Student name	_____
Course and section	_____
Repository name	_____
GitHub repository URL	_____
Date submitted	_____
Evidence PDF filename	_____
MS Teams submission status	_____

Quick Command Reference

```
git status
git add .
git commit -m "Describe the change"
git push
```